演1

$S_1 \xrightarrow{v_1} S_2 \xrightarrow{v_2} \phi$

1.

$v_1 = k_1[S_1]$

$v_2 = k_2[S_2]$

2.

$$\frac{d[S_2]}{dt} = v_1 - v_2$$

$$= k_1[S_1] - k_2[S_2]$$

演2

1.

$$\frac{d[Glu]}{dt} = -k[Glu]$$

$$\frac{d[Glu]}{[Glu]} = -k\,dt$$

$$\int \frac{d[Glu]}{[Glu]} = \int -k\,dt$$

$\log[Glu] = -kt + C$ (C:積分定数)

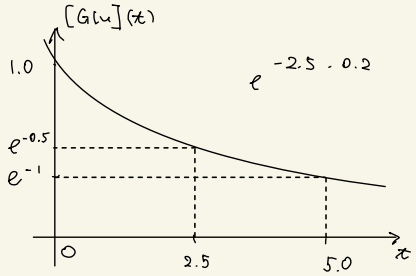
$[Glu] = e^{(-kt+C)}$

$= e^C \cdot e^{-kt}$

$t=0$ のとき $[Glu] = 1.0\,mM$ より、 $Glu = 1.0 \cdot e^{-kt}$

$= e^{-kt}$

2.



$e^{-0.2 \cdot 5}$

$e^{-2.5 \cdot 0.2}$

e^{-1}